Práctica Data WareHouse y SQL

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Representación gráfica de la distribución de estados por país.

Running Code

Se importa el CSV del resultado de la query.

```
df_flights <- read.csv("country_flight_status.csv")
```

Visualización del archivo

```
head(df_flights)
```

```
      country arrival_status count_flights min_delay max_delay average_delay
1      France             DY            23         5        45            16
2      France             OT             3         0         0             0
3      France         Unknow             1         0         0             0
4     Germany             OT             1         0         0             0
5       Italy             EY             1         3         3             3
6 Netherlands             DY             9         5        10             7
```

Podemos ver la información que contiene.

```
summary(df_flights)
```

```
   country          arrival_status     count_flights     min_delay
 Length:25          Length:25          Min.   : 1.00   Min.   :-15.00
 Class :character   Class :character   1st Qu.: 1.00   1st Qu.:  0.00
 Mode  :character   Mode  :character   Median : 6.00   Median :  0.00
                                       Mean   :10.64   Mean   :  2.68
                                       3rd Qu.:10.00   3rd Qu.:  5.00
                                       Max.   :67.00   Max.   : 24.00
   max_delay     average_delay
 Min.   : 0.00   Min.   :-2.00
 1st Qu.: 0.00   1st Qu.: 0.00
 Median : 0.00   Median : 0.00
 Mean   :11.28   Mean   : 6.28
 3rd Qu.:20.00   3rd Qu.:12.00
 Max.   :45.00   Max.   :25.00
```

Convertimos la columna country y arrival_status en factor.

```
df_flights$country <- as.factor(df_flights$country)
df_flights$arrival_status <- as.factor(df_flights$arrival_status)
summary(df_flights)
```

```
          country  arrival_status count_flights     min_delay
 Spain         :6   CX    :3       Min.   : 1.00   Min.   :-15.00
 Unknow        :6   DY    :6       1st Qu.: 1.00   1st Qu.:  0.00
 France        :3   EY    :3       Median : 6.00   Median :  0.00
 Netherlands   :3   NS    :2       Mean   :10.64   Mean   :  2.68
 United States :3   OT    :7       3rd Qu.:10.00   3rd Qu.:  5.00
 United Kingdom:2   Unknow:4       Max.   :67.00   Max.   : 24.00
 (Other)       :2
   max_delay     average_delay
 Min.   : 0.00   Min.   :-2.00
 1st Qu.: 0.00   1st Qu.: 0.00
 Median : 0.00   Median : 0.00
 Mean   :11.28   Mean   : 6.28
 3rd Qu.:20.00   3rd Qu.:12.00
 Max.   :45.00   Max.   :25.00
```

Cargamos la librerias de visualización

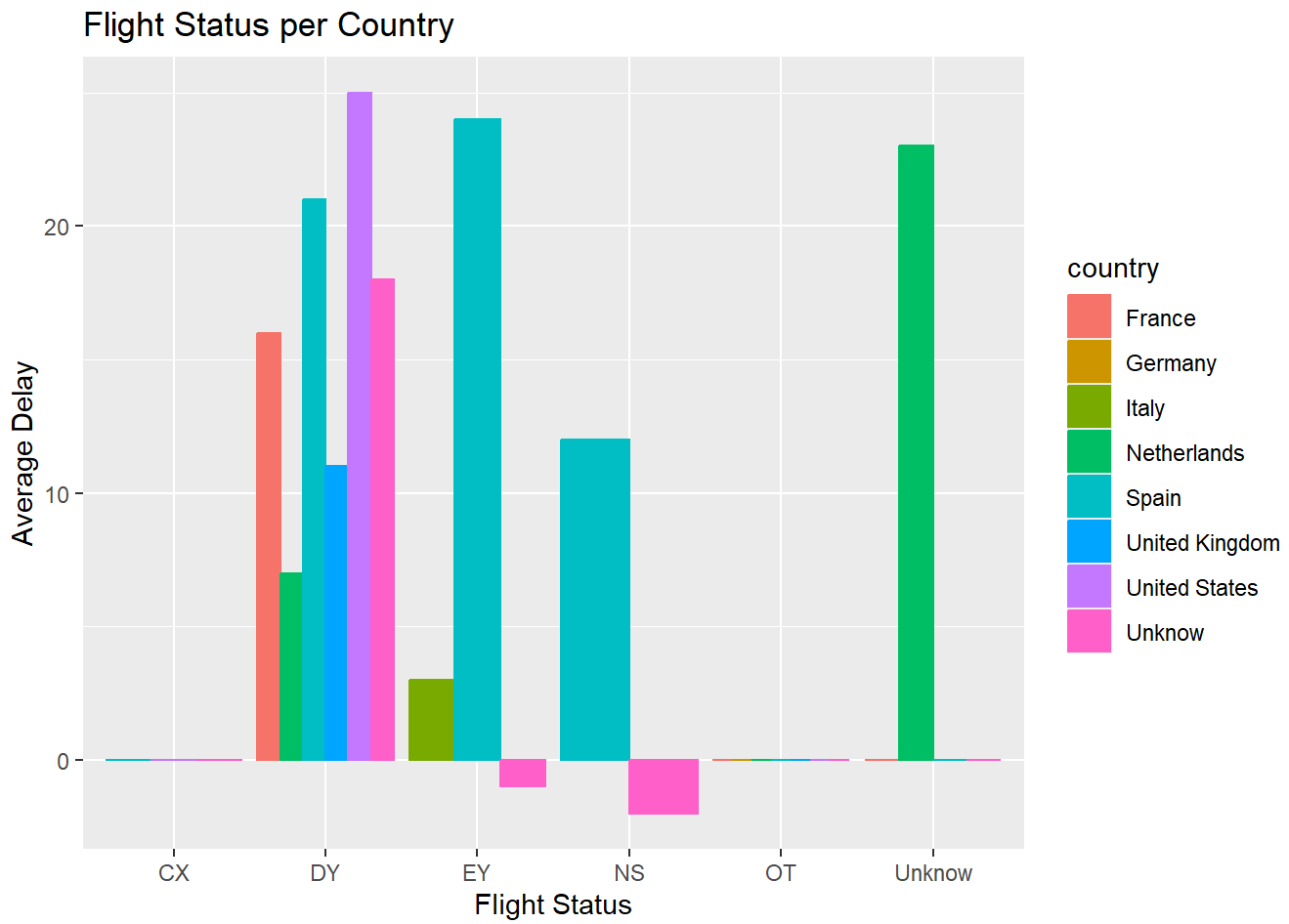
```
library("ggplot2")
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

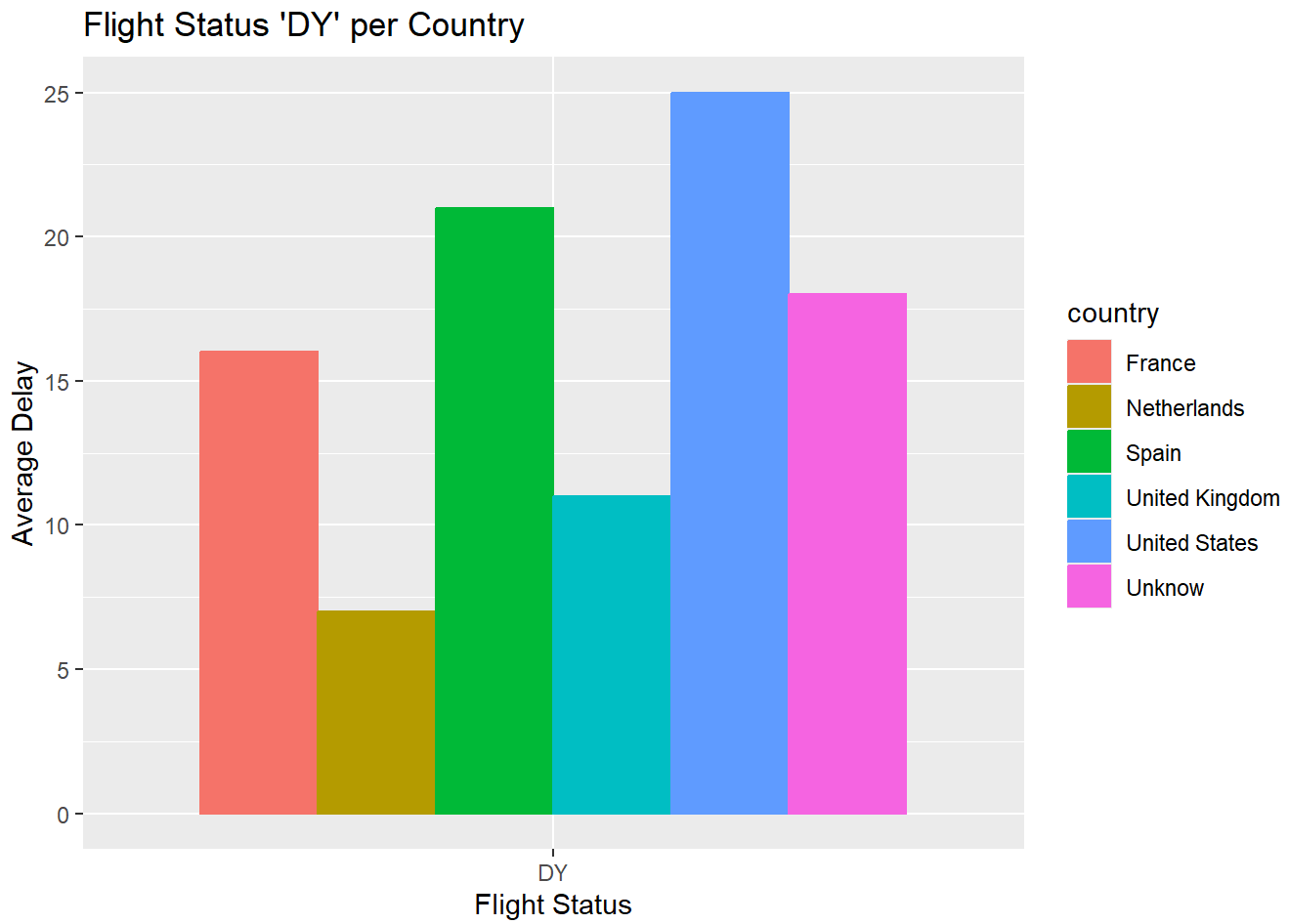
```
library(lubridate)
```

```
Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

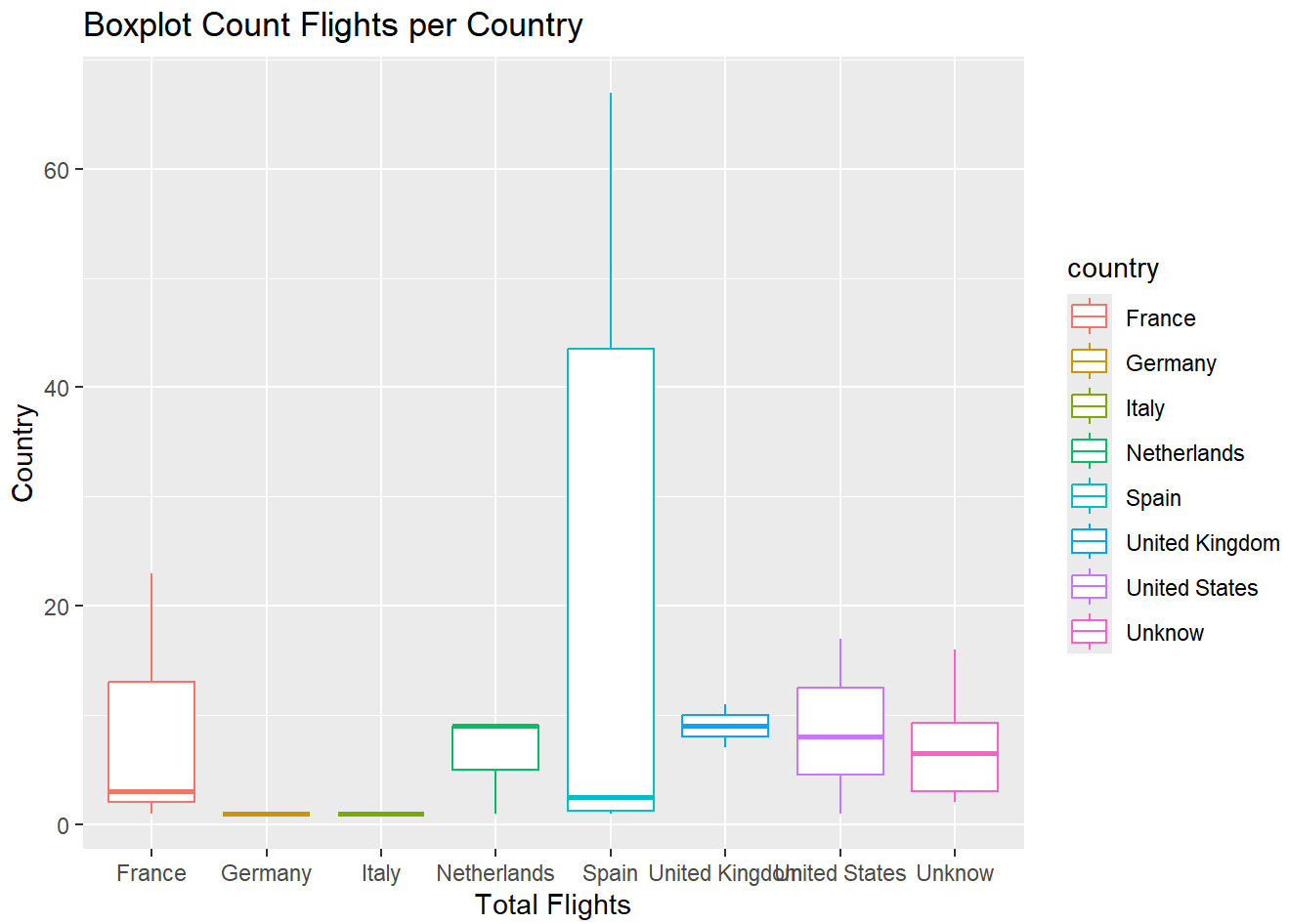
```
ggplot(df_flights, aes(x=arrival_status, color=country, fill=country, group=country)) + geom_col(
```




```
df_flights |> filter(arrival_status=='DY') -> df_flights_DY

ggplot(df_flights_DY, aes(x=arrival_status, color=country, fill=country, group=country)) + geom_c
```




```
ggplot(data=df_flights,aes(x=country,y=count_flights,color=country))+geom_boxplot()+ylab('Country
```



En el boxplot podemos identificar la distribución de la cantidad de vuelos por estado y pais.

```
# Filtramos Spain para ver que efectivamente la distribucion del boxplot es correcta.
df_flights |> filter(country=='Spain') -> df_flights_filtered
df_flights_filtered
```

```
  country arrival_status count_flights min_delay max_delay average_delay
1   Spain             CX             2         0         0             0
2   Spain             DY            67         5        45            21
3   Spain             EY             1        24        24            24
4   Spain             NS             1        12        12            12
5   Spain             OT            57         0         0             0
6   Spain         Unknow             3         0         0             0
```